

Privacy–Heterogeneity Trade-offs and Contribution Valuation in Federated Prostate Cancer Staging

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Abstract

Federated learning lets hospitals train a shared model without exchanging patient records, but its value depends on how privacy constraints, statistical heterogeneity and incentives interact. We study these interactions for pathologic staging (T3/T4 vs. T1/T2) of prostate cancer on TCGA-PRAD with a leakage-free pipeline (explicit feature whitelist, one sample per patient, train-only preprocessing) and 20 paired seeds per experiment. First, we replicate the strategy comparison of a recent hospital-participation study on the 9 real TCGA tissue-source sites: federated training gives hospitals no significant local gain over local models (FL–LOC = +0.015 [−0.007, +0.037]) but serves non-participating hospitals almost as well (free-rider penalty −0.008 [−0.013, −0.003]). Second, on the 347-patient clinical–protein cohort, Rényi-DP training costs little at $\varepsilon \geq 5$ but substantial utility at $\varepsilon = 1$ (paired loss −0.063 [−0.087, −0.038] AUC). FedProx reduces client drift by 32.0–36.4% and retains more utility under DP, but a step-size control shows that this gain is matched by halving FedAvg’s learning rate. Third, exact Shapley valuation shows that DP noise destabilises contribution rankings: with ten clients the Spearman correlation with the no-DP ranking falls from 0.73 at $\varepsilon = 10$ to 0.36 at $\varepsilon = 0.5$, and the top contributor keeps its place in only 0.30 of DP runs. Personalised fine-tuning improves on local training but not on the global model, proteomic features reduce rather than improve discrimination, and a confidence-threshold membership-inference attack is at chance level even without DP. All numbers are regenerated from code and checked by automated validation tests.

Keywords: Federated Learning, Differential Privacy, FedProx, Non-IID Learning, Prostate Cancer, Data Valuation, Shapley Value, Membership Inference.

1 Introduction

Pathologic staging of prostate cancer is vital for clinical risk stratification and treatment planning. Accurate identification of disease severity determines whether a patient undergoes radical prostatectomy, radiotherapy, active surveillance, or systemic androgen deprivation therapy. Machine learning models trained on combined clinical and molecular profiles can support clinicians in identifying high-risk disease. However, clinical and genomic cohorts are frequently isolated across institutions. Centralizing these datasets into a single repository is often impossible due to institutional barriers and data protection regulations (e.g., HIPAA and GDPR).

Federated Learning (FL) has emerged as a promising collaborative paradigm to address these limitations. Under FL, clinical sites cooperatively train a global model by sharing local update gradients or parameters rather than raw clinical records. Yet, clinical federated networks face three major challenges. First, statistical heterogeneity (non-IID data distributions) arising from variations in clinical protocols, regional demographics, and equipment calibration causes client update parameters to diverge from the global optimization trajectory, leading to client drift and suboptimal convergence. Second, the transfer of gradient updates is vulnerable to privacy reconstruction and membership leakage, requiring formal privacy guarantees such as Differential Privacy (DP). Third, participant sites require reliable contribution valuation mechanisms to justify collaboration, but naive metrics fail to capture marginal contributions.

Prior research has largely examined these challenges in isolation. The interactions between statistical heterogeneity, differential privacy noise, local personalization, and contribution valuation remain poorly understood. For ex-

ample, adding differential privacy noise to local updates disrupts optimization paths, and the severity of this disruption depends on the non-IID label skew across hospitals. Similarly, privacy noise alters computed game-theoretic contribution scores, potentially destabilizing incentive allocation.

To address these gaps, we systematically study the central research question: ****How do privacy constraints and statistical heterogeneity jointly affect federated clinical model utility, local adaptation, and participant contribution valuation?*** We construct a multi-dimensional conceptual model to characterize these relationships: clinical heterogeneity leads to client optimization drift, which is stabilized by FedProx regularization; this interaction occurs alongside privacy perturbation under sample-level differential privacy, establishing a privacy–utility trade-off; these combined constraints affect both global model utility and local personalized adaptation, which ultimately alters participant contribution valuation and its associated economic incentive allocation. The conceptual framework tracking these causal interactions is depicted in Fig. 1.

We characterise these interactions experimentally on TCGA-PRAD and first replicate the strategy comparison of Kazlouski *et al.* [17] on real hospital sites. Our main findings are:

- **Base-paper replication:** on 9 real TCGA sites, federated training brings no significant local gain over local models (FL–LOC = +0.015 [−0.007, +0.037]) while free-riders obtain nearly the same performance as participants (FR–FL = −0.008 [−0.013, −0.003]).
- **Drift containment (H1):** FedProx reduces client drift by 32.0–36.4% at every level of label skew; without DP this does not change accuracy.
- **Privacy–utility (H2):** utility is nearly unchanged at $\epsilon \geq 5$ but drops significantly at $\epsilon \leq 2$ (paired loss at $\epsilon = 1$: −0.063 [−0.087, −0.038]).
- **Joint constraints (H3):** under DP, FedProx beats FedAvg in 12 of 20 privacy–heterogeneity cells, but a FedAvg with half the learning rate matches it, so the gain is a step-size (noise-damping) effect.
- **Personalisation (H4):** proximal fine-tuning improves on local-only models (e.g. +0.051 [+0.015, +0.086] AUC for three clients) but not on the global model.
- **Valuation under privacy (H5):** DP noise destabilises exact Shapley rankings monotonically in ϵ ; in larger federations exact ranks and the top contributor are the least stable.
- **Empirical privacy audit (H7):** a confidence-threshold membership-inference attack is at chance even without DP for this model, so no empirical DP benefit can be shown.

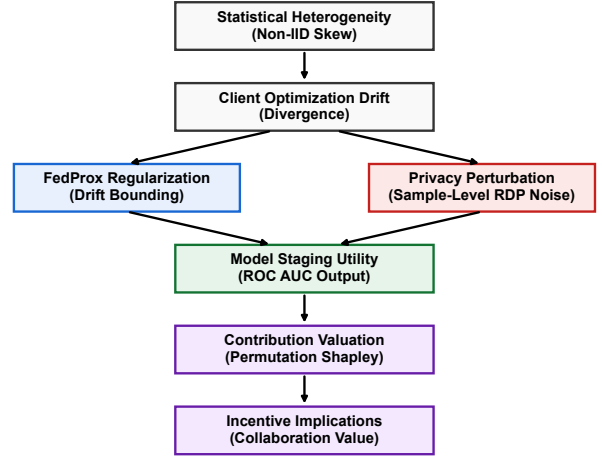


Figure 1: Conceptual model of the privacy–heterogeneity–optimization–valuation relationships investigated in this study, tracking variables from data skew to client optimization drift, proximal stabilization, privacy perturbation, staging utility, and collaborative incentive implications.

- **Feature evidence:** protein principal components reduce discrimination (−0.159 [−0.188, −0.130] AUC), and post-surgical pathology variables carry most of the signal.

The remainder of this manuscript is organized as follows: Section II reviews related work and defines the research gap. Section III details the system model and problem formulation. Section IV explains the mathematical methodology, including RDP accounting, Shapley valuation, and statistical analyses. Section V describes the dataset and preprocessing pipeline. Section VI outlines the experimental protocol. Section VII presents the empirical results. Section VIII discusses the survival extension, and Sections IX, X, XI, and XII present the discussion, limitations, future work, and conclusion.

2 Related Work

2.1 Federated Learning for Healthcare

Cross-silo federated learning has attracted significant interest in clinical AI due to its ability to train models across institutions without direct data sharing [19, 25, 26]. Clinical studies have applied FL to tasks such as brain tumor segmentation, medical imaging, rare cancer research, and tabular clinical risk prediction [24]. Recently, federated transfer learning has been applied to transfer prognostic models across related cancers under differential privacy [34], and decentralized networks have been developed to model pan-cancer survival using clinical-protein sequences [4]. To facilitate standardized implementations of these multi-modal models, benchmarks such as FedMultimodal have been established to provide unified recipes

for processing multi-site clinical and genomic features [9]. Tabular databases containing demographic variables, lab values, and genomic markers are highly sensitive, and institutional boundaries prevent their aggregation [33]. While early work assumed identical client data distributions, clinical datasets exhibit severe statistical skew due to differences in patient demographics, diagnostics, and regional pathology characteristics [14, 36].

2.2 Non-IID Optimization and Client Drift

When local datasets are non-IID, the local optimization objectives of clinical clients diverge. Standard Federated Averaging (FedAvg) aggregates updates by computing weighted averages of client parameters [21]. However, client drift causes local updates to guide parameters away from the global optimum, slowing convergence and reducing utility [31, 37]. To mitigate drift, FedProx adds a proximal regularization term to the local loss, penalizing deviations from the global model [18, 30]. Alternative approaches like SCAFFOLD use control variates to correct drift, but increase communication overhead by transmitting auxiliary parameters [15].

2.3 Differential Privacy and Rényi DP

Exposing model weights to a coordinator introduces privacy risks, as membership inference and model inversion attacks can reconstruct patient information. Differential Privacy (DP) provides mathematical guarantees against such leakage [6, 7]. DP-SGD applies gradient clipping to limit the influence of individual records, followed by Gaussian noise addition to perturb the aggregated update [1]. Tracking privacy leakage across communication rounds requires advanced composition methods. Rényi Differential Privacy (RDP) provides tight composition bounds for the Gaussian mechanism, making it suitable for multi-round federated training [22, 23]. Recent work has extended these mechanisms to handle across-silo user-level differential privacy, guaranteeing that an entire hospital’s dataset remains protected against aggregation leaks [16]. Furthermore, adaptive weight-based differential privacy (AWDP-FL) has been proposed to dynamically adjust gradient clipping thresholds based on neural network layers, minimizing the utility loss commonly associated with static clipping [5].

2.4 Client Contribution Valuation

To maintain the stability of clinical federated networks, contribution valuation methods are required to incentivize hospital participation [17, 32]. Leave-One-Out (LOO) methods assess a client’s marginal value by comparing global performance with and without that client [11]. However, LOO does not account for sub-coalition synergies and exhibits high variance. Game theory offers the Shapley value, which assigns contributions based on axiomatic

principles [10, 13]. However, computing exact Shapley values requires training models for all 2^K client coalitions, which is computationally expensive [12, 20, 28, 35]. A critical challenge arises when contribution valuation is performed under privacy constraints: standard i.i.d. noise injected for differential privacy can degrade the fidelity of estimated data values, rendering them highly unstable unless correlated noise injection is utilized [38]. To improve valuation efficiency, frameworks like VerFedSV have been proposed for vertical federated settings to assess feature contributions using intermediate outputs [8], while threshold-based KNN-Shapley (TKNN-Shapley) reduces complexity to linear time while natively supporting differential privacy guarantees [29].

2.5 Membership Inference Attacks in Clinical AI

While differential privacy provides formal mathematical bounds, empirical privacy audits are needed to measure practical resistance to leakage. Membership Inference Attacks (MIA) aim to determine whether a specific patient record was used in training [27]. In clinical applications, the prediction confidence difference between members and non-members serves as an information leak channel. Noise calibration under DP aims to obfuscate this confidence gap, reducing attacker advantage to random guess levels. Recent surveys categorize these attacks into loss-based, shadow-model, and decision-boundary strategies, analyzing diverse countermeasures across federated settings [2]. Moreover, advanced attacks like FedMIA have demonstrated that adversaries can exploit updates across multiple communication rounds to execute multi-stage inference attacks, highlighting the necessity of robust privacy audits beyond static confidence threshold evaluations [39].

2.6 Research Gap Analysis

Prior works have typically evaluated heterogeneity, differential privacy, personalization, and contribution valuation as isolated components. The multi-dimensional interactions between these constraints remain under-studied. For instance, the impact of DP noise on game-theoretic client rankings under heterogeneous skews has not been systematically characterized. This study addresses this gap. Table 1 contrasts our approach with existing literature.

3 System Model and Problem Formulation

Consider a federated network of K clinical clients and a central coordinator. Each client $k \in \{1, \dots, K\}$ holds a private dataset $D_k = \{(x_{i,k}, y_{i,k})\}_{i=1}^{n_k}$, where $x_{i,k} \in \mathbb{R}^d$ is a clinical feature vector and $y_{i,k} \in \{0, 1\}$ is the pathologic staging target ($1 = \text{T3/T4}$). Let $n = \sum_{k=1}^K n_k$.

Table 1: Comparison of Our Approach with Existing Federated Learning Literature

Study	Clinical FL	Non-IID Skew	FedProx	DP	PFL	Shapley	MIA	Privacy $\times$ Heterogeneity
McMahan et al. [21]	Yes	Yes	No	No	No	No	No	No
Li et al. [18]	Yes	Yes	Yes	No	No	No	No	No
Abadi et al. [1]	No	No	No	Yes	No	No	No	No
Ghorbani et al. [11]	No	No	No	No	No	Yes	No	No
Kazlouski et al. [17]	Yes	Yes (real sites)	No	No	No	No	No	No
Ours (DP-FPS)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The global objective is

$$\min_w F(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w), \quad (1)$$

where $F_k(w) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell(w; x_{i,k}, y_{i,k}) + \frac{\lambda}{2} \|w\|_2^2$. Under FedAvg, each client performs E local epochs of gradient descent from the current global model and the server averages the results:

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1}. \quad (2)$$

FedProx adds a proximal term to the local objective,

$$h_k(w; w^t) = F_k(w) + \frac{\mu}{2} \|w - w^t\|_2^2, \quad (3)$$

which limits how far local models move away from w^t ($\mu = 0$ recovers FedAvg).

Local gradients are clipped and perturbed with Gaussian noise calibrated by a Rényi-DP accountant. Client contributions are measured with the Shapley value of the utility $v(S)$, the test AUC of the FedAvg model trained on coalition S . The architecture is shown in Fig. 2.

4 Mathematical Methodology

4.1 Classification Model

Staging is modelled with logistic regression with an intercept,

$$p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}, \quad z_i = x_i^T w + b, \quad (4)$$

trained on binary cross-entropy $\ell(w; x_i, y_i) = -[y_i \ln p_i + (1 - y_i) \ln(1 - p_i)]$ with a small L_2 penalty ($\lambda = 10^{-3}$).

4.2 FedProx Local Optimization

At round t client k starts from w^t and performs E full-batch steps

$$w_k^{t,\tau+1} = w_k^{t,\tau} - \eta (\nabla F_k(w_k^{t,\tau}) + \mu(w_k^{t,\tau} - w^t)). \quad (5)$$

We measure client drift, for both FedAvg and FedProx, as

$$\bar{d} = \frac{1}{T} \sum_{t=1}^T \frac{1}{K} \sum_{k=1}^K \|w_k^t - w^{t-1}\|_2, \quad (6)$$

the distance between each local model after local training and the global model it started from.

4.3 Rényi Differential Privacy and Sensitivity Analysis

4.3.1 Privacy Unit and Adjacency

We use add/remove adjacency of single records, $D \sim D' \iff |D \Delta D'| = 1$. Because the analysis cohort keeps exactly one primary-tumour sample per patient (Section 5), a record is a patient and the guarantee is patient-level for this cohort.

Each per-sample gradient is clipped to L_2 norm C ,

$$\bar{g}_i = g_i / \max\left(1, \frac{\|g_i\|_2}{C}\right), \quad (7)$$

so adding or removing one patient changes the summed gradient by at most C . The released (noisy) gradient is

$$\tilde{g} = \frac{1}{n_k} \left(\sum_{i=1}^{n_k} \bar{g}_i + z \right), \quad z \sim \mathcal{N}(0, \sigma^2 C^2 I), \quad (8)$$

i.e. a Gaussian mechanism with noise multiplier σ on a sum query of sensitivity C (client sizes n_k are treated as public). The L_2 and proximal terms do not access data and are post-processing.

4.3.2 Rényi DP Composition

Updates are full-batch (sampling rate $q = 1$), so no sub-sampling amplification is claimed. A Gaussian mechanism with multiplier σ satisfies RDP of order $\alpha > 1$ with $\epsilon_{RDP}(\alpha) = \alpha/(2\sigma^2)$. Every patient participates in each of the $N = T \times E$ local steps of its own client, and clients hold disjoint patients, so

$$\epsilon = \min_{\alpha \in \mathcal{A}} \left\{ N \cdot \frac{\alpha}{2\sigma^2} + \frac{\ln(1/\delta)}{\alpha - 1} \right\}, \quad (9)$$

with a dense order grid $\mathcal{A}$ (1.1–4096). The accountant agrees with the closed-form optimum $\rho + 2\sqrt{\rho \ln(1/\delta)}$, $\rho = N/(2\sigma^2)$, to within 0.01% and is never below it. For $N = 100$ and $\delta = 10^{-5}$ the calibrated multipliers are $\sigma = 5.68, 10.55, 24.99, 49.02, 97.01$ for $\epsilon = 10, 5, 2, 1, 0.5$.

4.4 Exact Shapley Valuation

The Shapley value of client k is

$$\phi_k = \sum_{S \subseteq \mathcal{N} \setminus \{k\}} \frac{|S|!(K - |S| - 1)!}{K!} [v(S \cup \{k\}) - v(S)], \quad (10)$$

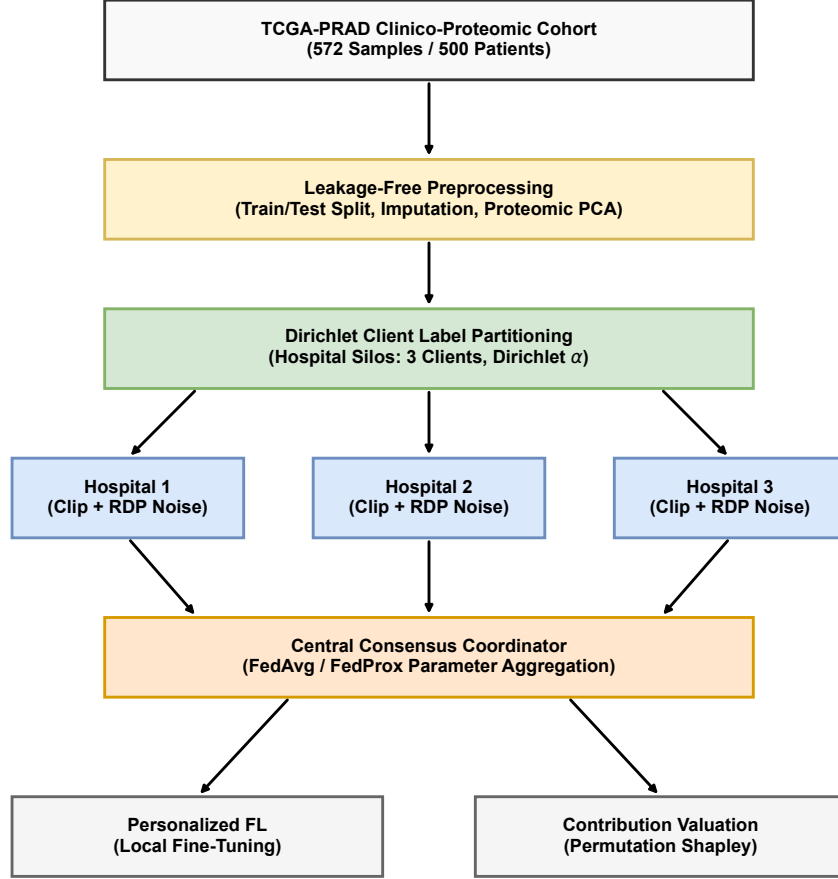


Figure 2: Experimental system architecture of the DP-FPS clinical federated system: client-level preprocessing, local gradient clipping and RDP-calibrated noise, FedProx aggregation, and post-training Shapley valuation.

with $v(\emptyset) = 0.5$. We compute it *exactly* by training all $2^K - 1$ coalitions (vectorised across coalitions; numerically identical to separate training runs) for $K \leq 10$, so no Monte-Carlo error confounds the effect of DP noise. Leave-one-out (LOO) is $v(\mathcal{N}) - v(\mathcal{N} \setminus \{k\})$. Under DP every coalition run draws independent noise.

4.4.1 Ranking Stability Metrics

For each partition and DP noise realisation r we compare the DP Shapley ranking with the no-DP ranking of the same partition using Spearman’s ρ , the rank-reversal rate $RR = \frac{1}{K} \sum_k \mathbb{I}(\text{rank}_k^{DP} \neq \text{rank}_k^{noDP})$, and the top-contributor stability

$$TS = \frac{1}{R} \sum_{r=1}^R \mathbb{I}(\arg \max_k \phi_{k,r}^{DP} = \arg \max_k \phi_k^{noDP}), \quad (11)$$

whose chance level is $1/K$. (An earlier version averaged $\mathbb{I}(\text{rank}_{k,r}^{DP} = 1)$ over clients, which equals $1/K$ by construction for every configuration.)

4.5 Membership Inference Attack Formulation

We audit empirical leakage with a confidence-threshold membership-inference attack. The score of a record is the model’s confidence in its true label, $s_i = y_i p_i + (1 - y_i)(1 - p_i)$. Members (training patients) are subsampled to the number of non-members (test patients) and the attack is summarised by its ROC AUC. As a positive control, the same attack is applied to a deliberately over-fitted model (200 random noise features, no regularisation).

4.6 Domain Shift Formulations

Robustness is probed with synthetic perturbations of the held-out split: covariate shift scales each numeric feature by $\beta_j \sim U(1 - 0.25s, 1 + 0.25s)$ and adds $\mathcal{N}(0, (0.5s)^2)$ noise; concept shift flips a fraction $\min(0.25s, 0.5)$ of labels, for severity s .

4.7 Statistical Analysis

Every experiment is repeated over 20 seeds. A seed re-draws the stratified 80/20 split, the client partition, model noise and DP noise, so seed-to-seed variation covers all sources of randomness. We report means with t -based 95% confidence intervals over seeds. Comparisons between methods are *paired* by seed; we report the mean paired difference with its 95% CI and a two-sided Wilcoxon signed-rank test, and call a difference significant only if its CI excludes zero. For ranking stability we additionally apply a one-sided Wilcoxon test of $d_i = \rho_i - 0.90 < 0$, where ρ_i is the seed-level Spearman correlation. Per-client AUCs are averaged only over clients whose local test split contains both classes (macro AUC); we do not pool predictions of different local models, because pooling rewards each model for learning its own client’s base rate.

5 Datasets and Preprocessing

5.1 TCGA-PRAD Cohorts

We use the clinical, survival and RPPA protein data of TCGA-PRAD [3]. The clinical file contains 572 sample records from 500 patients, including normal-tissue and metastatic samples. We keep primary-tumour samples (barcode type -01) with a recorded pathologic T stage and one sample per patient, giving a *clinical cohort* of 493 patients (305 T3/T4). Restricting to patients with protein data (352 samples) gives the *matched cohort* of 347 patients (237 T3/T4), used for all DP-FPS experiments so that clinical-only and multimodal models are compared on identical patients. Each seed splits it 80/20 into 277 training and 70 test patients.

5.2 Feature Whitelist and Leakage Control

Features are taken from an explicit whitelist: age, race, ethnicity, clinical T and M stage, prior malignancy (pre-operative), plus primary and secondary Gleason pattern, Gleason sum and pathologic N stage (post-surgical pathology). After one-hot encoding (reference level dropped) this gives 18 features (13 pre-operative). Identifiers (sample, case, treatment and pathology-report IDs), record timestamps, and outcome or follow-up variables (vital status, days to last follow-up, treatment fields) are excluded; an earlier version of the pipeline admitted about 1,650 one-hot columns built from such per-patient identifiers. Gleason pattern and pathologic N come from the same prostatectomy report as the pathologic T label, so they are strong but label-adjacent predictors; we therefore also report a pre-operative-only model. Imputation (median, or an explicit *Unknown* level), scaling and encoding are fitted on training rows only.

For the multimodal ablation, the 487 RPPA targets are filtered to the 456 with at most 30% missing values, median-imputed, standardised and reduced by PCA

to 95% explained variance, all fitted on the training split (about 115 components, depending on the split; 129 if PCA were fitted on all 347 patients).

5.3 Client Partitions

Real sites. For the replication of the base paper we use the TCGA tissue-source site as the hospital: the 9 sites with at least 20 patients (384 patients) are clients, each split 80/20 (stratified when possible); the 109 patients of smaller sites never train and act as external free-riders. Preprocessing statistics are shared across sites (as obtainable by secure aggregation).

Dirichlet label skew. For DP-FPS we draw, for each class, client proportions from $\text{Dir}(\alpha)$ and apply the *same* proportions to the training and the test split, so every client has a local test set with its own label skew; $\alpha \in \{100, 10, 1, 0.5, 0.1\}$ (smaller = more heterogeneous). Unless stated otherwise $K = 3$.

5.4 Survival Events

The matched cohort contains only 9 deaths (8 in the training and 1 in the test split for the reference seed; 12 deaths over all 572 records). Survival modelling is therefore not evaluated.

6 Experimental Protocol

Unless stated otherwise: $T = 20$ rounds, $E = 5$ local epochs ($N = 100$ steps), learning rate $\eta = 0.5$, $\lambda = 10^{-3}$, $\mu = 0.5$, $C = 1$, $\delta = 10^{-5}$, 20 seeds (Table 2). The optimiser settings were chosen by cross-validation of the centralised NumPy model, whose AUC then matches scikit-learn’s logistic regression (0.818 vs 0.824, Table 5).

Table 2: Experimental Protocol

Parameter	Value
Raw records / patients	572 / 500
Clinical cohort (1 sample/patient)	493
Matched cohort (train/test)	347 (277/70)
Clinical features (full / pre-op)	18 / 13
Proteins (ablation)	487 $\rightarrow$ 456 $\rightarrow$ PCA 95%
Rounds T / local epochs E	20 / 5
Learning rate η / L_2 λ	0.5 / 10^{-3}
FedProx μ	0.5
Clipping C / δ	1.0 / 10^{-5}
Seeds per experiment	20

7 Results

7.1 Replication of the Base Paper on Real Sites

Table 4 and Fig. 3 replicate the strategy comparison of Kazlouski *et al.* [17] on the 9 TCGA sites. Averaged

Table 3: Claim–evidence summary. Verdicts are computed from the result files: ‘supported’ only if the 95% CI of the paired difference over seeds excludes 0 in the claimed direction.

Area	Claim	Evidence (mean [95% CI])	Verdict
Base paper	Federated model beats each hospital’s local model (per-site AUC)	FL - LOC = +0.015 [-0.007, +0.037]	not significant
Base paper	Free-riders get (almost) the same benefit as participants	FR - FL = -0.008 [-0.013, -0.003] (negative = small free-rider penalty)	supported
Base paper	Federated training approaches centralised (pooled) training	CEN - FL = -0.001 [-0.004, +0.002]	supported
Base paper	Learned model beats a simple rule baseline (Gleason sum)	FL - rule = +0.067 [+0.048, +0.086]	supported
Our model	Adding proteomics (PCA) improves staging over clinical features	(full+protein) - full = -0.159 [-0.188, -0.130] over 25 CV folds	opposite
Our model	H1 FedProx reduces client drift under heterogeneity	relative drift reduction ($\alpha=100$: 32.0%, $\alpha=10$: 33.2%, $\alpha=1$: 35.6%, $\alpha=0.5$: 36.0%, $\alpha=0.1$: 36.4%); lowest CI bound 31.4%	supported
Our model	FedProx improves accuracy without DP ($\alpha = 0.1$)	FedProx - FedAvg = +0.002 [-0.001, +0.004]	not significant
Our model	H2 Strong privacy ($\epsilon = 1$) costs utility	FedAvg($\epsilon=1$) - FedAvg(no DP) = -0.063 [-0.087, -0.038]	supported
Our model	Weak privacy ($\epsilon = 10$) costs little utility (< 0.03 AUC)	FedAvg($\epsilon=10$) - no DP = -0.002 [-0.009, +0.005]	supported
Our model	H3 FedProx beats FedAvg under joint DP + heterogeneity	CI > 0 in 12/20 (α, ϵ) cells vs FedAvg; in 3/20 cells vs a FedAvg with half the learning rate (step-size control)	explained by step size
Our model	H4a Personalised FL beats local-only training	PFL - Local (K=3, macro client AUC) = +0.051 [+0.015, +0.086]	supported
Our model	H4b Personalised FL beats the global FedProx model	PFL - FedProx = -0.006 [-0.022, +0.011]	not significant
Our model	H5 DP noise destabilises Shapley contribution rankings	K=10: Spearman vs no-DP 0.73 ($\epsilon=10$) $\rightarrow$ 0.36 ($\epsilon=0.5$); top-contributor kept 0.62 $\rightarrow$ 0.30	supported
Our model	Shapley and LOO give similar client rankings	Spearman(Shapley, LOO): K=3 0.68, K=10 0.44	partly
Our model	H7 DP reduces membership-inference vulnerability	attack AUC no-DP 0.511 [0.489, 0.533] vs $\epsilon=1$ 0.510; positive control (over-fitted) 0.659	not significant

over sites with a defined AUC (7.4 per seed), local training reaches 0.775, FedAvg 0.790, free-riding 0.782, centralised training 0.789 and the Gleason-sum rule 0.723. The gain of joining the federation over training locally is not significant (FL–LOC = +0.015 [−0.007, +0.037]), FedAvg matches centralised training (CEN–FL = −0.001 [−0.004, +0.002]), and the free-rider penalty is small (FR–FL = −0.008 [−0.013, −0.003]). These reproduce the base paper’s central observations: hospitals gain little *locally* from participating, while a federated model serves non-participating hospitals almost as well. The learning curve (Fig. 4) shows free-rider AUC rising from 0.751 with one training site to 0.785 with all 9, with most of the gain reached after four to five sites. All learned strategies clearly beat the training-free rule (FL–BL = +0.067 [+0.048, +0.086]). LOO contributions are close to zero for every site, whereas exact Shapley values are positive and distinguishable; the two rankings agree only moderately (Spearman $\rho = 0.48$ [0.34, 0.62]).

7.2 Centralised Baselines and Feature Sets

Table 5 compares feature sets with repeated cross-validation. The full clinical set reaches 0.818 with the NumPy model used in federation. Adding the protein principal components *lowers* discrimination (paired difference −0.159 [−0.188, −0.130]), and protein features alone reach only 0.574: with roughly 115 components for 277 training patients the linear model overfits. Removing the post-surgical pathology features costs

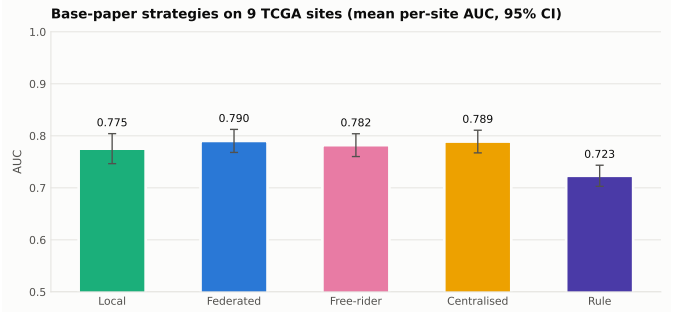


Figure 3: Base-paper strategies on the TCGA sites: mean per-site test AUC with 95% CI over 20 seeds.

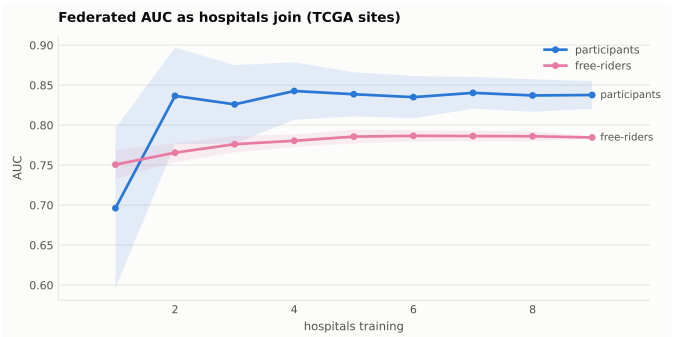


Figure 4: FedAvg AUC for participating sites and for free-riders (non-participating sites and the external small-site pool) as sites join in random order.

Table 4: Base-paper replication on the 9 TCGA tissue-source sites with ≥ 20 patients: mean per-site test AUC over 20 seeds (LOC = local only, FL = FedAvg over all sites, FR = free-rider, CEN = pooled, BL = Gleason-sum rule) and mean contribution (exact Shapley, leave-one-out). – = AUC undefined (single-class test split).

Site	n	T3/T4	LOC	FL	FR	CEN	BL	Shapley	LOO
Pittsburgh	95	0.59	0.843	0.824	0.795	0.825	0.682	+0.048	+0.0058
IGC	65	0.40	0.741	0.771	0.772	0.764	0.682	+0.045	+0.0005
MD Anderson	45	0.73	0.900	0.914	0.911	0.914	0.779	+0.045	-0.0002
Roswell Park	43	0.60	0.698	0.735	0.722	0.735	0.710	+0.034	-0.0010
Indivumed	31	0.68	0.623	0.632	0.632	0.632	0.670	+0.029	-0.0006
UCSF	31	0.68	0.750	0.745	0.745	0.742	0.660	+0.027	+0.0026
PROCURE	27	1.00	–	–	–	–	–	+0.037	-0.0008
Hamburg	24	0.96	1.000	1.000	1.000	1.000	1.000	+0.038	-0.0001
Lahey Clinic	23	0.39	0.783	0.825	0.808	0.825	0.775	+0.035	+0.0004
Macro mean	384	–	0.775	0.790	0.782	0.789	0.723		

+0.176 [+0.148, +0.203] (pre-operative model 0.642). The remaining experiments therefore use the full clinical feature set; the multimodal and pre-operative variants are reported in the ablation.

7.3 Client Drift under Statistical Heterogeneity (H1)

Table 6 and Fig. 5 report client drift (Eq. 6). Drift grows with heterogeneity (FedAvg: 0.206 at $\alpha = 100$ to 0.770 at $\alpha = 0.1$), and FedProx reduces it by 32.0–36.4% at every α , with 95% CIs far from zero; increasing μ from 0 to 1 lowers drift monotonically (0.599 to 0.268 at $\alpha = 0.5$). H1 is supported. The reduction in drift does not translate into higher test AUC without DP: no paired AUC difference is significant, because the convex logistic objective lets FedAvg reach a good solution regardless.

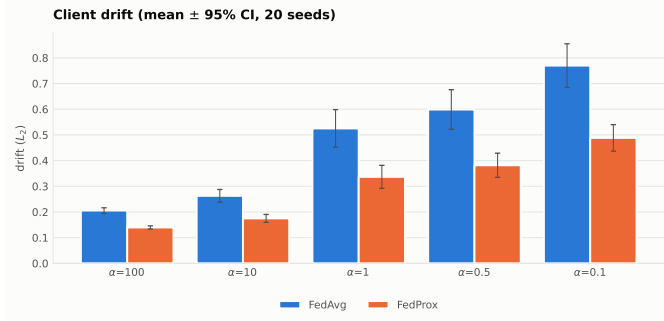


Figure 5: Client drift (Eq. 6) of FedAvg and FedProx across Dirichlet α (mean and 95% CI over 20 seeds).

7.4 Privacy–Utility Trade-off (H2)

Fig. 6 shows global test AUC against the privacy budget at $\alpha = 10$. Without DP FedAvg reaches 0.823. Weak privacy is almost free ($\epsilon = 10$: 0.821, paired loss -0.002 [$-0.009, +0.005$]), whereas strong privacy is costly ($\epsilon = 2$: -0.023 [$-0.038, -0.008$]; $\epsilon = 1$: -0.063 [$-0.087, -0.038$]; $\epsilon = 0.5$:

-0.132 [$-0.180, -0.085$]). With 20 paired seeds the degradation at $\epsilon \leq 2$ is significant, supporting H2.

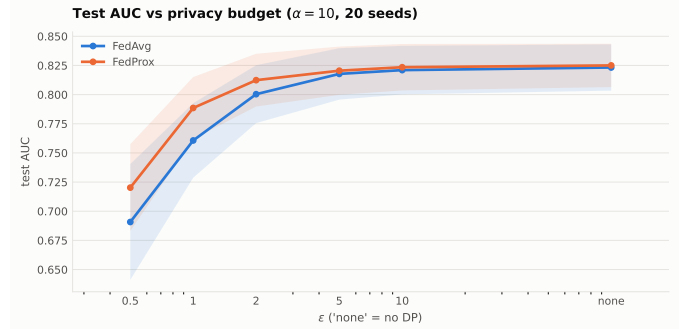


Figure 6: Global test AUC versus privacy budget ϵ ($\alpha = 10$, $K = 3$; mean and 95% CI over 20 seeds).

7.5 FedProx under Joint Privacy and Heterogeneity (H3)

Table 7 and Fig. 7 cross label skew with privacy. Without DP, FedProx and FedAvg are indistinguishable (largest mean difference 0.0019). Under DP, FedProx is significantly better in 12 of 20 cells, by 0.028–0.030 AUC at $\epsilon \in \{1, 0.5\}$, largely independent of α . A step-size control explains this: a FedAvg run with half the learning rate matches FedProx (FedProx better in only 3 of 20 cells; largest mean difference 0.003 at $\epsilon \leq 1$). Under DP the proximal term acts mainly as a smaller effective step that damps the injected noise, rather than as a correction for heterogeneity. H3 is therefore supported only in this qualified sense.

7.6 Personalised Adaptation (H4)

Table 8 and Fig. 8 evaluate each client on its own test split ($\alpha = 0.5$). With three clients, FedProx reaches 0.811 per client against 0.754 for local-only models, and personalised fine-tuning (PFL)

Table 5: Centralised baselines: mean AUC over repeated stratified cross-validation (25 folds, 5×5) on the 347-patient cohort. All preprocessing is fitted inside each training fold.

Model	Clinical (full)	Clinical (pre-op)	Protein only	Full + protein	Pre-op + protein
NumPy LR (ours)	0.818	0.642	0.574	0.659	0.595
LR (L2)	0.824	0.661	0.633	0.682	0.638
LR (L1)	0.831	0.662	0.637	0.712	0.659
Random forest	0.803	0.644	0.599	0.782	0.627
MLP	0.792	0.670	0.599	0.664	0.623

Table 6: Client drift $\bar{d} = \text{mean}_{t,k} \|w_k^t - w^{t-1}\|_2$ without DP (3 clients, 20 seeds, mean $\pm$ sd). Spread = max-min T3/T4 rate across clients. Last column: paired FedProx–FedAvg test-AUC difference with 95% CI.

α	Spread	FedAvg drift	FedProx drift	Reduction % [95% CI]	ΔAUC [95% CI]
100	0.07	0.206 ± 0.023	0.140 ± 0.013	32.0 [31.4, 32.5]	$+0.0022 [-0.0012, +0.0057]$
10	0.20	0.263 ± 0.053	0.175 ± 0.033	33.2 [32.6, 33.8]	$+0.0018 [-0.0013, +0.0049]$
1	0.55	0.525 ± 0.157	0.337 ± 0.096	35.6 [34.9, 36.2]	$+0.0019 [-0.0009, +0.0046]$
0.5	0.68	0.599 ± 0.165	0.382 ± 0.100	36.0 [35.5, 36.5]	$+0.0016 [-0.0024, +0.0056]$
0.1	0.84	0.770 ± 0.181	0.488 ± 0.110	36.4 [35.9, 36.9]	$+0.0018 [-0.0009, +0.0044]$

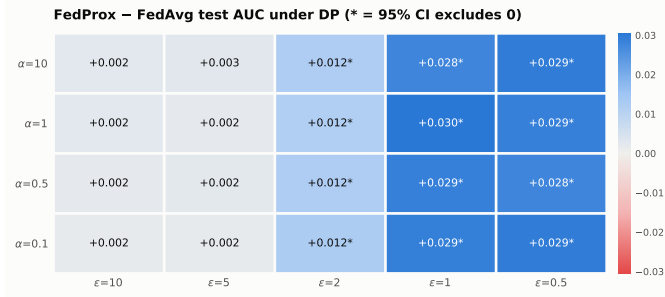


Figure 7: Paired FedProx–FedAvg test-AUC difference across α and ε (20 seeds; * = 95% CI excludes 0).

reaches 0.805 (PFL–Local = $+0.051 [+0.015, +0.086]$). PFL does not improve on the global FedProx model (PFL–FedProx = $-0.006 [-0.022, +0.011]$ for $K = 3$, $-0.003 [-0.014, +0.007]$ for $K = 10$). With $K = 10$ the comparison is again in favour of collaboration (PFL–Local = $+0.053 [+0.010, +0.096]$). Because many clients have a single-class test split under strong skew, these averages cover only the clients with a defined AUC (Table 8). H4 is supported for PFL versus local training, but not for PFL versus the global model.

7.7 Contribution Valuation under Privacy Noise (H5)

Without DP, exact Shapley values are positive for every client and track client size only moderately (Spearman with size: 0.55 [0.31, 0.79] for $K = 3$, 0.46 [0.35, 0.58] for $K = 10$); LOO rankings agree with Shapley rankings only partially (0.68 [0.42, 0.93] for $K = 3$, 0.44 [0.30, 0.58] for $K = 10$), because LOO ignores sub-coalitions and becomes nearly zero when one client is redundant.

Table 9 and Figs. 9–10 quantify how DP changes these

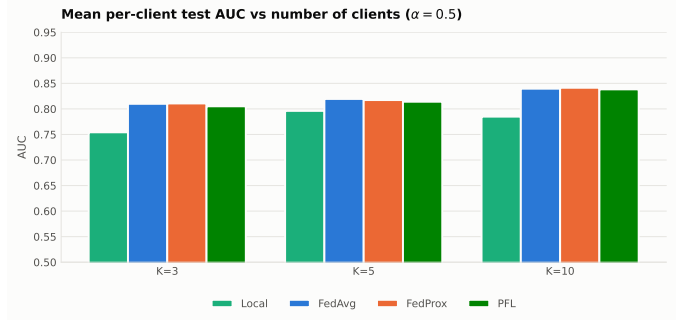


Figure 8: Mean per-client test AUC of local, FedAvg, FedProx and personalised models for $K \in \{3, 5, 10\}$ clients ($\alpha = 0.5$, 20 seeds).

values. Each DP realisation is compared with the no-DP ranking of the same partition, so the Shapley computation itself contributes no randomness (exact enumeration). Rank agreement falls monotonically as privacy strengthens: for $K = 3$, Spearman ρ drops from 0.68 at $\varepsilon = 10$ to 0.25 at $\varepsilon = 0.5$, and for $K = 10$ from 0.73 to 0.36. Even the weakest budget leaves ρ well below 0.90 (one-sided Wilcoxon $p < 0.01$ in every configuration). The probability that the no-DP top contributor is still ranked first falls from 0.79 to 0.51 for $K = 3$ (chance 0.33) and from 0.62 to 0.30 for $K = 10$ (chance 0.10), and at $\varepsilon = 0.5$ the rank-reversal rate reaches 0.88 for $K = 10$. The overall rank correlation degrades by a similar amount for every federation size, whereas exact ranks and the top contributor are least stable in larger federations. H5 is supported: an incentive scheme that pays hospitals according to Shapley values computed from DP-trained models would frequently reward the wrong hospital.

Table 7: Global test AUC under joint label skew (α) and privacy (ϵ), 3 clients, mean over 20 seeds. FedAvg ($\eta/2$) is a step-size control. Δ = paired FedProx–FedAvg difference; bold = 95% CI excludes 0.

	Model	no DP	$\epsilon = 10$	$\epsilon = 5$	$\epsilon = 2$	$\epsilon = 1$	$\epsilon = 0.5$
$\alpha = 10$	FedAvg	0.823	0.821	0.818	0.800	0.761	0.691
	FedProx	0.825	0.823	0.821	0.812	0.789	0.720
	FedAvg ($\eta/2$)	0.826	0.821	0.820	0.812	0.786	0.718
	Δ Prox–Avg	–	+0.002	+0.003	+0.012	+0.028	+0.029
$\alpha = 1$	FedAvg	0.823	0.821	0.818	0.801	0.757	0.688
	FedProx	0.825	0.823	0.820	0.812	0.787	0.717
	FedAvg ($\eta/2$)	0.825	0.821	0.820	0.812	0.786	0.715
	Δ Prox–Avg	–	+0.002	+0.002	+0.012	+0.030	+0.029
$\alpha = 0.5$	FedAvg	0.824	0.821	0.818	0.799	0.758	0.687
	FedProx	0.826	0.823	0.820	0.811	0.787	0.715
	FedAvg ($\eta/2$)	0.826	0.821	0.820	0.811	0.784	0.715
	Δ Prox–Avg	–	+0.002	+0.002	+0.012	+0.029	+0.028
$\alpha = 0.1$	FedAvg	0.825	0.821	0.818	0.796	0.753	0.684
	FedProx	0.826	0.823	0.820	0.809	0.782	0.713
	FedAvg ($\eta/2$)	0.827	0.821	0.820	0.810	0.782	0.712
	Δ Prox–Avg	–	+0.002	+0.002	+0.012	+0.029	+0.029

Table 8: Mean per-client test AUC ($\alpha = 0.5$, 20 seeds) over clients whose local test split contains both classes (#def. = mean number of such clients). PFL = proximal fine-tuning of the FedProx model. Paired differences with 95% CI.

K	#def.	Local	FedAvg	FedProx	PFL	PFL–Local	PFL–FedProx
3	2.2	0.754	0.810	0.811	0.805	+0.051 [+0.015, +0.086]	–0.006 [–0.022, +0.011]
5	2.9	0.796	0.820	0.817	0.814	+0.018 [–0.021, +0.057]	–0.003 [–0.011, +0.005]
10	3.5	0.785	0.840	0.841	0.838	+0.053 [+0.010, +0.096]	–0.003 [–0.014, +0.007]

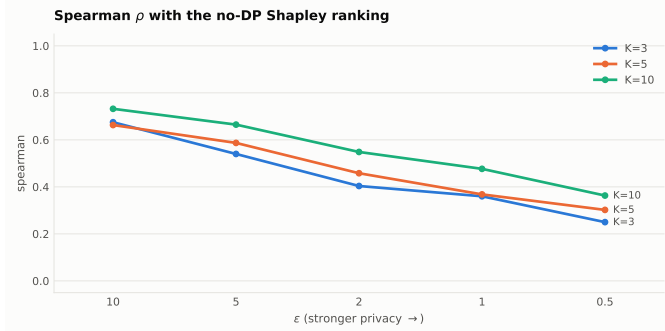


Figure 9: Spearman correlation between DP and no-DP exact Shapley rankings versus ϵ for $K \in \{3, 5, 10\}$ clients (20 partitions).

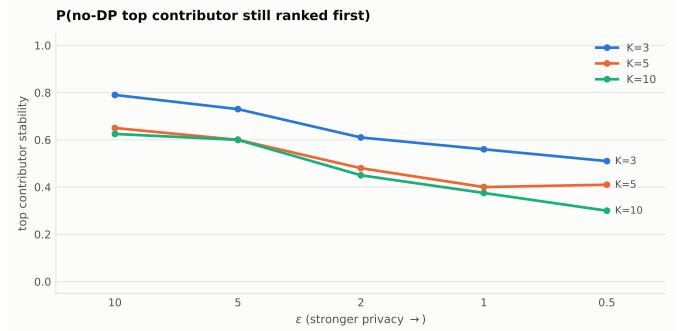


Figure 10: Top-contributor stability (probability that the no-DP top contributor stays first under DP) versus ϵ ; chance level is $1/K$.

7.8 Empirical Membership-Inference Evaluation (H7)

Without DP the confidence-threshold attack is at chance level (attack AUC 0.511 [0.489, 0.533]), and it stays there under DP ($\epsilon = 1$: 0.510 [0.484, 0.537]; $\epsilon = 0.5$: 0.512 [0.490, 0.534]). The positive control confirms that the attack works when a model memorises its training data (AUC 0.659 [0.640, 0.678]). The 18-feature logistic model generalises well (train–test AUC gap +0.017 [–0.007, +0.041]), so there is little membership

signal for DP to remove. The formal DP guarantee still holds, but its *empirical* benefit cannot be demonstrated with this attack, and H7 is not supported. (The previously reported reduction from 0.568 to 0.480 arose in a pipeline whose identifier features let the model memorise individual patients.)

Table 9: Stability of exact Shapley rankings under DP relative to the no-DP ranking (20 partitions; 5, 5, 2 noise realisations per partition for $K = 3, 5, 10$). TS = probability that the no-DP top contributor is still ranked first (chance = $1/K$). p : one-sided Wilcoxon test of $\rho - 0.90 < 0$.

K	ϵ	Spearman ρ [95% CI]	Rank reversal	TS	Chance TS	p
3	10	0.68 [0.50, 0.85]	0.30	0.79	0.33	0.007
3	5	0.54 [0.35, 0.73]	0.38	0.73	0.33	4.1e-4
3	2	0.40 [0.21, 0.60]	0.48	0.61	0.33	1.4e-4
3	1	0.36 [0.16, 0.56]	0.51	0.56	0.33	6.4e-5
3	0.5	0.25 [0.07, 0.43]	0.57	0.51	0.33	4.3e-5
5	10	0.66 [0.55, 0.78]	0.52	0.65	0.20	7.0e-5
5	5	0.59 [0.45, 0.72]	0.59	0.60	0.20	5.1e-5
5	2	0.46 [0.30, 0.62]	0.66	0.48	0.20	4.4e-5
5	1	0.37 [0.21, 0.52]	0.73	0.40	0.20	4.4e-5
5	0.5	0.30 [0.15, 0.45]	0.74	0.41	0.20	4.4e-5
10	10	0.73 [0.67, 0.79]	0.76	0.62	0.10	5.2e-5
10	5	0.66 [0.58, 0.75]	0.77	0.60	0.10	9.5e-7
10	2	0.55 [0.45, 0.64]	0.79	0.45	0.10	4.4e-5
10	1	0.48 [0.39, 0.56]	0.82	0.38	0.10	4.4e-5
10	0.5	0.36 [0.26, 0.47]	0.88	0.30	0.10	4.4e-5

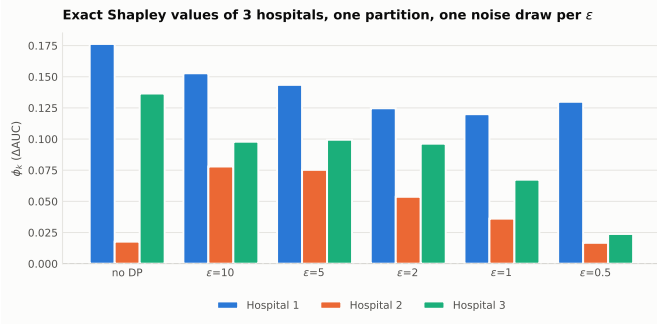


Figure 11: Exact Shapley values of three clients for one partition: no DP and one DP noise draw per ϵ .

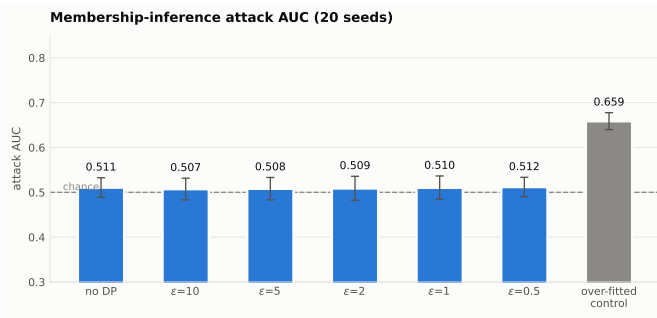


Figure 12: Membership-inference attack AUC (mean and 95% CI over 20 seeds) without DP, under DP, and for an over-fitted positive control. 0.5 = chance.

7.9 Controlled Distribution-Shift Stress Test (H6)

Under synthetic covariate shift the FedProx model degrades gracefully (AUC 0.826 $\rightarrow$ 0.765 at severity 2), while concept shift, which flips labels, drives it to 0.498 (Fig. 13).

This is a sensitivity analysis on perturbed copies of the test split, not an external validation.

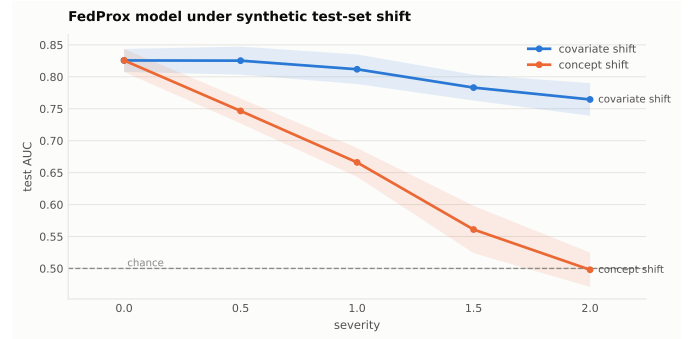


Figure 13: FedProx test AUC under synthetic covariate and concept shift of the test split (20 seeds).

7.10 Ablation

Table 10 summarises the components on the global test split ($K = 3$, $\alpha = 0.5$). FedAvg (0.824) and FedProx (0.826) match centralised training (0.823). DP costs little at $\epsilon = 5$ (0.820) and more at $\epsilon = 1$ (0.787). Using only pre-operative features (0.659) or adding protein components (0.682) both reduce discrimination substantially.

8 Survival Analysis Limitation

A federated Cox model was implemented, but the matched cohort contains only 9 deaths among 347 patients (1 in the reference test split), so out-of-sample survival discrimination cannot be estimated reliably. No survival claims are made.

Table 10: Ablation on the global test split ($K = 3$, $\alpha = 0.5$, 20 seeds, mean $\pm$ sd; paired difference to FedProx with 95% CI).

Configuration	Test AUC	vs FedProx
Centralised (pooled data)	0.823 ± 0.042	$-0.003 [-0.008, +0.002]$
FedAvg	0.824 ± 0.041	$-0.002 [-0.006, +0.002]$
FedProx	0.826 ± 0.038	reference
FedProx + DP ($\epsilon=5$)	0.820 ± 0.045	$-0.006 [-0.012, +0.001]$
FedProx + DP ($\epsilon=1$)	0.787 ± 0.058	$-0.039 [-0.056, -0.022]$
FedProx: pre-op features only	0.659 ± 0.062	$-0.167 [-0.193, -0.141]$
FedProx: clinical (full) + protein PCs	0.682 ± 0.050	$-0.144 [-0.166, -0.122]$

9 Discussion

Four results stand out. First, replicating the base paper on real TCGA sites reproduces its sustainability argument: joining a federation brings a hospital no significant local gain over its own model, yet a federated model serves non-participating hospitals almost as well as participants, and most of the free-rider benefit is reached with four to five contributing sites. Second, privacy has a clear utility price below $\epsilon \approx 2$, and FedProx softens it, but a step-size control shows that the benefit comes from damping noisy local steps rather than from correcting heterogeneity; without DP, FedProx reduces client drift by about a third without changing accuracy. Third, DP noise severely destabilises Shapley-based contribution rankings at every federation size, which matters for any payment or reputation scheme built on valuations of privately trained models. Fourth, the empirical privacy audit is uninformative for this low-capacity model: membership inference is at chance even without DP.

The corrected pipeline also changes two substantive conclusions of earlier drafts. Proteomic principal components do not improve staging on this cohort; they reduce AUC. And much of the predictive signal comes from post-surgical pathology variables that are recorded together with the pathologic T label, so the clinically realistic pre-operative task is considerably harder (AUC 0.642).

9.1 Threats to Validity

9.1.1 Internal Validity

Dirichlet clients are simulated; the real-site analysis uses TCGA tissue-source sites, which are biobanks rather than independent clinical deployments, and their test splits are small (5–19 patients), so single-site AUCs are noisy and only averages over seeds should be interpreted. Hyperparameters were selected by cross-validation on the same cohort. Exact Shapley values were computed for up to 10 clients; larger federations would require approximation.

9.1.2 External Validity

The evaluation is retrospective and limited to TCGA-PRAD. The strongest predictors (Gleason pattern, pathologic N) are post-surgical; a pre-operative tool would reach

only about 0.642 AUC with the available variables. PSA, a key pre-biopsy predictor in the base paper, is not available in TCGA.

9.1.3 Privacy Validity

The guarantee is per patient for the analysed cohort (one sample per patient), assumes full-batch updates ($q = 1$), and treats client sizes and the shared preprocessing statistics as public. The empirical audit uses one confidence-threshold attack and does not cover gradient-reconstruction or model-inversion attacks.

9.1.4 Statistical Validity

All comparisons are paired over 20 seeds; seed-level replicates share the same 347 patients, so the CIs describe variability across splits and partitions rather than across independent cohorts. Cross-validation fold differences are correlated, so their CIs are indicative.

9.2 Reproducibility Statement

All results, tables and figures are regenerated from code: `python run_all.py` recomputes every experiment into `results/final/`, `python -m fl_study.validate` runs 22 automated checks (cohort counts, absence of identifier features, train-only preprocessing, RDP accountant against the closed form, empirical noise scale, clipping bound, Shapley axioms, equivalence of vectorised and per-coalition training, and reproducibility), and `python Paper/generate_paper_assets.py` writes every number in this manuscript. Code: https://github.com/JeyanthPonnaluri/Mini_Project_RVR.

10 Future Work

Future work will evaluate patient-level DP with subsampling amplification, noise-robust valuation (e.g. valuation on a separate, non-private validation channel), approximate Shapley values for larger federations, and validation on independent multi-institutional cohorts that include pre-operative PSA and enough survival events.

11 Conclusion

We studied privacy, heterogeneity, personalisation and contribution valuation in federated prostate-cancer staging on TCGA-PRAD with a leakage-free pipeline, exact Shapley values and 20 paired seeds per experiment. On real TCGA sites, federated training gives hospitals no significant local gain but serves free-riders almost as well as participants, reproducing the base paper’s sustainability argument. Rényi-DP training costs little utility at $\epsilon \geq 5$ and substantially more at $\epsilon \leq 1$; FedProx reduces client drift by 32.0–36.4% and preserves accuracy under DP, but the latter effect is matched by halving FedAvg’s

learning rate. Personalised fine-tuning improves on local training but not on the global model. Most importantly, DP noise markedly destabilises Shapley contribution rankings (Spearman ρ with the no-DP ranking falls to 0.36 at $\varepsilon = 0.5$ with ten clients), so contribution-based incentives in privacy-preserving federations need noise-robust valuation.

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